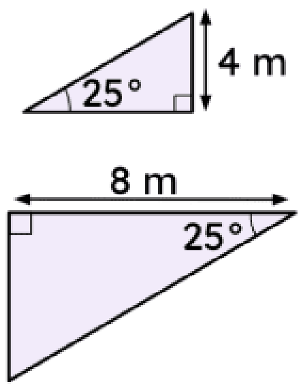
Problem solving with trigonometry

1 In a right triangle, if the shortest sides are 48 cm and 20 cm, what is the length of the hypotenuse? Tick **1** correct answer

- [x] 52 cm
- [ ] 50 cm
- [ ] 54 cm

2 Which of the following statements is true for these triangles? Tick **1** correct answer



- [x] The triangles are similar.
- [ ] The triangles are not similar.
- [ ] It is not possible to know whether the triangles are similar or not.

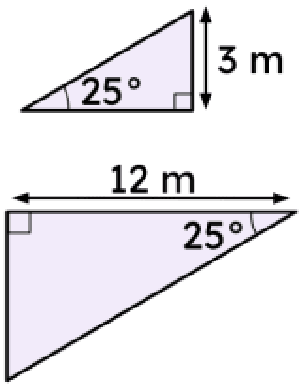
3 In a right triangle, if the hypotenuse is 45 cm and a second side is 36 cm, what is the length of the third side? (Use a calculator to help you.) Tick **1** correct answer

- [ ] 28 cm
- [x] 27 cm
- [ ] 26 cm

4 In a right triangle, if the hypotenuse is 15 cm and a second side is 12 cm, what is the perimeter? (Use a calculator to help you.) Tick **1** correct answer

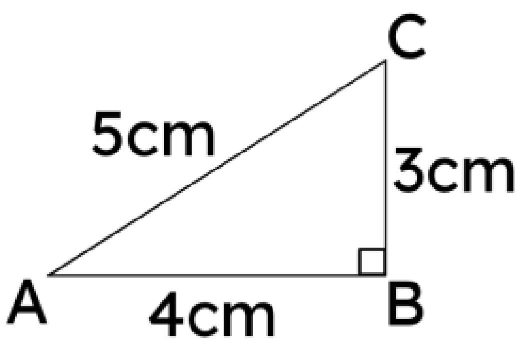
- [ ] 34 cm
- [x] 36 cm
- [ ] 32 cm

5 For this pair of triangles, can you determine whether they are similar without using side lengths? Tick **1** correct answer



- [x] Yes, because their three angles correspond.
- [ ] No because you only know two angles.
- [ ] No, because you always need a side and two angles.

6 Could an equilateral triangle be similar to the triangle below? Tick **1** correct answer



- [ ] Yes
- [x] No